

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AMA580
Subject Title	Advanced Topics in InsurTech
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	None
Objectives	This course aims to equip students with a deep understanding of emerging trends, tools, and innovations in the rapidly evolving field of InsurTech. By exploring advanced topics such as blockchain applications, big data, regulatory technology (RegTech), and the integration of artificial intelligence in insurance, students will develop the analytical and technical skills necessary to navigate and lead in the InsurTech landscape. The course emphasises critical thinking, hands-on experience, and strategic insight to prepare students for impactful roles in insurance, technology, and entrepreneurship.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Analyse and Evaluate advanced concepts and emerging technologies in the InsurTech ecosystem, like blockchain, artificial intelligence, machine learning, and big data analytics in insurance industry and decision-making. b) Apply quantitative and computational techniques to solve complex problems in risk management, and financial modelling. c) Assess regulatory frameworks and ethical considerations relevant to InsurTech innovations, including RegTech and data privacy. d) Design and prototype InsurTech solutions using modern development tools and platforms. e) Communicate technical and strategic insights effectively to both technical and nontechnical stakeholders through project presentation.
Subject Synopsis/ Indicative Syllabus	Digital Transformation in Insurance: Explore AI, chatbots, cloud computing, and big data reshaping the insurance industry. Blockchain and InsurTech Distribution: Learn how blockchain is reinventing insurance and how digital platforms are changing distribution models. IoT and Parametric Insurance: Understand how wearable devices and real-time data enable personalised insurance solutions. InsurTech Showcase & Guest Talks: Demonstrations of cutting-edge tech and insights from industry professionals. Startups, Regulation & Risk Management: Examine InsurTech ecosystems, legal frameworks, data protection, and RegTech innovations. Experience learning through practical projects: Students will engage in immersive learning through hands-on labs that utilize industrial

	software to tackle technical challenges or gain firsthand insights during visits to leading FinTech companies, and conclude the course by showcasing their final projects in presentations followed by peer feedback and interactive discussions.																																								
Teaching/Learning Methodology	The subject will be delivered primarily through lectures, with a strong emphasis on the latest developments in InsurTech. The teaching and learning approach are designed to foster a deep understanding of how InsurTech concepts could be applied to address real-world challenges in professional settings. Students are encouraged to engage in deep learning by employing advanced cognitive strategies such as critical analysis, evaluative thinking, and the integration of theory with practice. This approach aims to cultivate the ability to apply theoretical knowledge to practical financial technology problems encountered in everyday work life. This course is designed to immerse students in practical InsurTech applications through industry guest lectures, real-world case showcases, visits to tech firms and hands-on experience with industrial software.																																								
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. In-class Exercises/ Discussion</td><td>10%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>2. Mid-term Test</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>3. Final Presentation</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100 %</td><td colspan="5"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The In-class Exercises / Discussion requires students to participate in interactive exercises and discussions to reinforce key concepts, apply theory to real-world scenarios, and develop critical thinking and collaboration skills in a dynamic, practical learning environment. The Midterm Test consists of a series of questions based on the material covered in the course. It is designed to assess students’ understanding of key concepts, practical techniques, and problem-solving skills related to insurance technology development. The Final Presentation requires students to explore and present a specific topic, demonstrating their grasp of InsurTech concepts and their ability to apply these concepts in practical contexts. It will be used to assess students’ understanding of InsurTech key concepts learned in this course. The oral presentation and presentation slide reflect students’ reflective thinking and effective communication skills.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	1. In-class Exercises/ Discussion	10%	✓	✓	✓	✓		2. Mid-term Test	40%	✓	✓	✓	✓		3. Final Presentation	50%	✓	✓	✓	✓	✓	Total	100 %					
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Student Study Effort Expected	Class contact:	
	▪ Lecture	39 Hrs.
	Other student study effort:	
	▪ Readings and exercises	20 Hrs.
	▪ Final Presentation Preparation	20 Hrs.
	▪ Self-study	30 Hrs
	Total student study effort	109 Hrs.
Reading List and References	Reference will base on the latest internet / case studies related to the above topics.	

March 2026